

ICT Intermarket Strategy — Session Handover

Date: 2026-05-20 **From:** Claude Code on the Web session sourced from `thabisocollinsengane/pulsify` **To:** New Claude Code session sourced from `ict-intermarket-strategy` repo

0. What you are picking up

A fully-scaffolded ICT day-trading algorithm written in Python for QuantConnect (LEAN engine). The repo contains 13 files, 952 lines of code, two clean commits on `main`. The strategy compiles in principle but has **not yet been backtested** — that's the next session's first job.

The strategy trades **GBPUSD** and **EURUSD** using ICT concepts (FVGs, Order Blocks, liquidity sweeps, killzones) gated by intermarket analysis (synthetic DXY + EURGBP relative strength) with a ForexFactory news filter. Designed for fully autonomous execution on QuantConnect cloud connected to OANDA — no manual intervention, since the user's trading psychology demands hands-off operation.

1. Decisions log

Decision	Choice	Rationale
Platform	QuantConnect (Python, cloud)	Free, browser/mobile, real autonomous live trading via OANDA, best multi-symbol & news support
Language	Python	QC's primary supported language
Pairs traded	GBPUSD, EURUSD only	User's chosen scope
Confirmation pairs	DXY (synthetic) + EURGBP	Per user spec — DXY = USD direction, EURGBP = pair selection
DXY source	Synthetic via ICE formula	OANDA on QC doesn't expose DXY directly
Killzones	London Open, NY AM, London Close (NY time)	User selected these three; Asia excluded
HTF bias	4H + 1H must agree with signal direction	Standard ICT layered confirmation
LTF setup	15m sweep + 5m FVG	Classic ICT silver-bullet style entry refinement

Decision	Choice	Rationale
Min target	20 pips	Hard user requirement
Min R:R	2:1	Default; tunable
Target sources	FVG / OB / equal high or low on H4, Daily, Weekly	Per user spec; nearest valid wins
Pyramiding	Add on continuation FVG, max 3 legs	Each new leg promotes prior leg's SL to break-even
Risk per leg	1% of equity	Standard; tunable in <code>config.py</code>
News filter	ForexFactory weekly XML, ±15 min around H/M USD/EUR/GBP events	Free source, downloadable from algo via <code>self.Download()</code>
Execution	Fully autonomous on QC live	User explicitly asked for hands-off (psychology)
Repo host	New private GitHub repo <code>ict-intermarket-strategy</code>	User created it; new session sourced from there

2. Strategy specification (the rules)

Entry trigger — checked on every 5m bar inside a killzone

1. **Killzone check:** is the current UTC time inside London Open / NY AM / London Close (NY time)? If not, exit. Also exit if we're in the last 15 minutes of a killzone (`NO_NEW_TRADES_LAST_MIN`).
2. **News check:** is `self.UtcTime` within ±15 min of a High or Medium impact USD/EUR/GBP event? If yes, exit.
3. **DXY 1H bias:** compute synthetic DXY from 6 constituents; run BOS-based bias detection. Must be +1 or -1, not 0.
4. **EURGBP 1H bias:** same BOS-based bias detection.
5. **Resolve intermarket signal** via the 4-row decision table (see §3). If null, exit. If the resolved pair is not the pair we're evaluating, exit.
6. **HTF pair agreement:** 4H and 1H bias on the *traded* pair must both match the signal's direction. If either disagrees, exit.
7. **15m liquidity sweep:** the most recent 15m bar must have taken out a prior 20-bar high (for shorts) or low (for longs) AND closed back inside the range.
8. **5m FVG:** the most recent three 5m bars must form a Fair Value Gap in the trade direction, with gap ≥ 3 pips.
9. **HTF target:** scan H4, Daily, Weekly for nearest *unmitigated* FVG, OB, or equal-high/low in trade direction. If none found, exit.
10. **20-pip filter:** if distance from FVG-mid to nearest target < 20 pips, exit.

11. **R:R check:** stop = sweep extreme ± 1 pip; entry = FVG midpoint; reward:risk must be ≥ 2.0 .
Else exit.
12. **Place orders:** Limit @ FVG mid, Stop-market @ sweep ± 1 pip, Limit (TP) @ HTF target.

Pyramiding — checked on every 5m bar when a position is open

1. `MAX_LEGS = 3` total (initial + 2 adds). If at cap, exit.
2. News check (same rule).
3. New continuation FVG in trade direction must exist on 5m.
4. Current price must be ≥ 10 pips in favour of the *previous leg's entry*.
5. Reward from new entry to existing target must still be ≥ 20 pips.
6. New leg's stop = previous leg's entry → promotes previous leg to break-even.
7. Place orders identically (limit entry + SL + TP at shared HTF target).

Intermarket decision table

DXY bias	EURGBP bias	Decision	Logic
Bear	Bull	LONG EURUSD	USD weak, EUR is the strong leg → buy it
Bear	Bear	LONG GBPUSD	USD weak, GBP is the strong leg → buy it
Bull	Bull	SHORT GBPUSD	USD strong, GBP is the weak leg → sell it
Bull	Bear	SHORT EURUSD	USD strong, EUR is the weak leg → sell it
Neutral on either side	—	No trade	Confirmation incomplete

3. Architecture — file by file

```

ict-intermarket-strategy/
├── main.py                327 lines — QCAAlgorithm entry point
├── config.py              44 lines — every tunable parameter
├── intermarket.py         32 lines — DXY×EURGBP decision table
├── news_filter.py         55 lines — ForexFactory XML parser + block checker
├── risk.py                44 lines — position sizing + pyramiding TradeState
├── ict/
│   ├── __init__.py
│   ├── killzones.py       41 lines — NY-time gating
│   ├── bias.py            42 lines — BOS-based HTF bias
│   ├── fvg.py             72 lines — FVG detect, mitigate, nearest unmitigated
│   ├── order_block.py     70 lines — OB detect + mitigate
│   ├── liquidity.py       80 lines — equal H/L clustering + sweep detection
│   └── dxy_synthetic.py   35 lines — ICE formula synthetic DXY
├── README.md             110 lines — quickstart + config + caveats
└── .gitignore

```

Module responsibilities

- **main.py** — the only file QC sees as the algorithm class. Wires up symbol subscriptions, consolidators (5m/15m/1H/4H/Daily/Weekly), handles `OnData`, drives `_maybe_open` and `_maybe_pyramid`, manages target search across H4/D/W, cleans up state in `OnOrderEvent`.
- **config.py** — all parameters. Touch this first when iterating.
- **intermarket.py** — pure function `resolve(dxy_bias, eurgbp_bias) -> IntermarketSignal | None`. No state.
- **news_filter.py** — `NewsCalendar` class with `.load(xml_text)` and `.is_blocked(utc_dt)`. Stateless after load. Refreshed weekly via QC `Schedule`.
- **risk.py** — `position_size(equity, entry, stop, symbol)` and `TradeState` dataclass tracking legs.
- **ict/killzones.py** — `can_open_new_trade(utc_dt)` is the gate.
- **ict/bias.py** — `htf_bias(candles)` returns +1/-1/0.
- **ict/fvg.py** — `detect_new_fvg(candles, symbol)` checks the latest 3 bars.
- **ict/order_block.py** — scans for OBs with mitigation marking.
- **ict/liquidity.py** — equal H/L clustering and `detect_sweep` (last-bar based).
- **ict/dxy_synthetic.py** — ICE formula: $50.14348112 * \text{EURUSD}^{-0.576} * \text{USDJPY}^{0.136} * \text{GBPUSD}^{-0.119} * \text{USDCAD}^{0.091} * \text{USDSEK}^{0.042} * \text{USDCHF}^{0.036}$.

4. Config defaults and rationale (`config.py`)

Parameter	Default	Why
<code>STARTING_CASH</code>	10,000	Reasonable backtest baseline
<code>RISK_PER_TRADE_PCT</code>	1.0	Industry standard for systematic trading
<code>MAX_LEGS</code>	3	User spec
<code>MIN_PIPS_TARGET</code>	20	User hard requirement
<code>MIN_RR</code>	2.0	Conservative; tune after first backtest
<code>KILLZONES</code>	London Open / NY AM / London Close (NY time)	User selected
<code>NO_NEW_TRADES_LAST_MIN</code>	15	Avoid stuck trades into killzone close
<code>NEWS_BLOCK_MINUTES_BEFORE/ AFTER</code>	15 / 15	Standard window; tune up for NFP if needed
<code>NEWS_IMPACTS</code>	High, Medium	Per user spec
<code>NEWS_CURRENCIES</code>	USD, EUR, GBP	Only currencies relevant to GBPUSD/EURUSD
<code>LTF_ENTRY_RES_MIN</code>	5	5m execution
<code>LTF_SETUP_RES_MIN</code>	15	15m sweep detection
<code>HTF_BIAS_RES_MIN</code>	60	1H bias
<code>TARGET_TF_MINUTES</code>	(240, 1440, 10080)	H4, Daily, Weekly target scan
<code>SWING_LOOKBACK</code>	20	Bars for BOS swing definition
<code>EQ_HIGH_LOW_TOLERANCE_PIPS</code>	2	Tightness of "equal" clustering
<code>FVG_MIN_SIZE_PIPS</code>	3	Filter out micro-FVGs
<code>OB_LOOKBACK_BARS</code>	200	OB scan horizon

5. Known gaps and honest caveats

1. **Not yet backtested.** Compiles in principle but has never run. Expect to find QC API mistakes on first run.

2. **Synthetic DXY** is a close ICE approximation, not the official ICE DXY. High/low computation uses per-bar highs/lows of each constituent, which doesn't truly map to DXY's intra-bar extremes — it's a bias proxy, not a precise level.
 3. **_targets_in_series** recomputes FVGs from scratch each call. Performance optimisation deferred — re-evaluate after first backtest profile.
 4. **Pyramiding moves prior-leg SL to BE only at the moment a new leg fires.** No intermediate trailing.
 5. **OANDA brokerage model hardcoded** in `main.py` via `SetBrokerageModel(BrokerageName.OandaBrokerage, ...)`. Change if going to a different broker.
 6. **No walk-forward / parameter optimisation.** Defaults baked in.
 7. **News parsing depends on ForexFactory XML schema stability.** If FF changes the format, `news_filter.py` parser breaks silently (returns 0 events). Log the count.
 8. **News refresh runs once on Monday at 00:05 UTC.** If the algorithm starts mid-week, only the initial load (`Initialize`) has events; the first scheduled refresh comes the next Monday.
 9. **Target-search timeframes (Daily, Weekly) need long warmup.** `SetWarmUp(timedelta(days=20))` may not give enough Weekly bars to find unmitigated targets. May need to bump to 60–90 days.
 10. **No equity-curve circuit breaker.** A bad day can lose multiple legs across both pairs simultaneously.
-

6. First-session priorities (what the new session should do, in order)

1. **Read README.md and main.py end-to-end.** Build a mental model.
 2. **Run a code review pass** — flag QC API mistakes, type errors, anything that won't compile. Particular suspects: - `RollingWindow[TradeBar](300)` syntax in Python QC - `TradeBarConsolidator` event subscription pattern - `self.Download()` exception behaviour - `LimitOrder` returning a ticket vs. the bracket pattern used (separate SL + TP orders)
 3. **First backtest:** ~6 months of data, GBPUSD + EURUSD, default config. Capture: total trades, win rate, profit factor, max drawdown, average R, news-block count.
 4. **Iterate on visible issues** — adjust warmup, sweep lookback, FVG min-size, etc.
 5. **Compare with/without intermarket gating** — ON vs. OFF; profit factor should improve with gating. If it doesn't, the intermarket model is wrong and we revisit.
 6. **Then forward-test on OANDA demo** for 2–4 weeks before any real money.
-

7. Things the user explicitly does NOT want

- Manual intervention. The algorithm must analyse and execute autonomously.
 - A computer-required workflow. Everything has to run in the cloud and be controllable from mobile/browser.
 - Trading outside the three chosen killzones.
 - Trading during high/medium impact USD/EUR/GBP news windows.
 - Targets less than 20 pips away.
 - More than 3 legs on any single trade.
-

8. Things the user did NOT specify (use sensible defaults, ask if uncertain)

- Backtest period — defaulted to 2024-01-01 to 2025-04-30 in `main.py`.
 - Starting capital — defaulted to \$10,000.
 - Max trades per day — not capped (only natural cap is "one open position per pair at a time"). Consider proposing a daily cap after first backtest.
 - Stop-trading-after-N-losses circuit breaker — not implemented.
 - Notifications (Telegram / Discord / email) — not implemented.
-

9. Kickoff prompt for the new session

Paste the following verbatim into the new session sourced from the ICT repo:

I'm continuing work on this ICT intermarket trading strategy from a previous session. Please read `HANDOVER.md` first (it's at the repo root) — it contains the full strategy spec, design decisions, known gaps, and first-session priorities. Then read `README.md` and `main.py`.

Your first task: run a thorough code review of the scaffold. Flag any QuantConnect API mistakes, Python type errors, logic bugs, or things that won't compile cleanly. Don't fix anything yet — just produce a numbered list of issues sorted by severity (critical → cosmetic) so I can decide what to tackle first. Be especially suspicious of: - `RollingWindow[TradeBar]` syntax and consolidator wiring - `self.Download()` exception handling - The bracket-order pattern (separate limit entry + stop-market SL + limit TP) - `OnOrderEvent` cleanup logic when one of multiple SL/TP orders fills - The synthetic DXY high/low calculation in `_dxy_bias_1h`

After the review, recommend the smallest set of fixes that unblocks a first backtest on QuantConnect. Then propose a backtest configuration (date range, initial capital, expected trade frequency) and we'll iterate from there.

10. Useful links

- **QuantConnect docs:** <https://www.quantconnect.com/docs/v2/writing-algorithms>
 - **QC LEAN GitHub:** <https://github.com/QuantConnect/Lean>
 - **OANDA REST API:** <https://developer.oanda.com/rest-live-v20/introduction/>
 - **ForexFactory weekly XML:** https://nfs.faireconomy.media/ff_calendar_thisweek.xml
 - **ICT educational content:** Inner Circle Trader YouTube (free)
-

End of handover. Good luck with the first backtest.